

Dr. Chao YIN

☎ +86 150 9602 8758 | ✉ cyinac@connect.ust.hk / chaoyin@gdas.ac.cn | 🌐 GitHub | 🏠 G Scholar | 📍 Room 202, No. 108-1, Compound 81, Yuexiu District, Guangzhou, China

💡 RESEARCH INTERESTS

3D point cloud processing; **foundation models** (SAM, DINO, CLIP); **LLMs** and **multi-modal 2D/3D LLMs** (LLaVA, LISA); semantic/instance segmentation, weakly-supervised and long-tailed learning; large-scale dataset construction; **remote sensing (RS)** and **GIS** applications; AI-augmented research workflows.

★ RESEARCH HIGHLIGHTS

- Expertise in **3D computer vision and deep learning**: point cloud segmentation, 3D scene understanding, long-tailed learning, weakly-supervised, and unsupervised learning for urban, construction, and industrial applications.
- Strong experience in **multi-modal AI systems**: integrating 2D images, 3D LiDAR, and other modality data for automated scene analysis and intelligent decision-making in complex environments.
- Open-source contributor and practical coder**: Developed and maintained **6+ influential repositories** on GitHub including ResPointNet++ (41 stars), Deep-Learning-On-Point-Clouds (62 stars), SE-PseudoGrid, and SQN implementations.
- Pioneer in **AI-augmented research methodology**: leveraging LLMs (Claude, GPT-5, GLM 4.6) for rapid prototyping, literature synthesis, and research workflow optimization while maintaining rigorous scientific standards.
- Strong publication and funding track record**: Published **3 first-author Q1 papers** with **7 papers under review/in progress**; PI of **680K RMB** from China Postdoctoral Science Foundation and Guangdong Province Overseas Postdoctoral Talent Program.
- Foundation Models & Multi-modal LLMs**: Hands-on with SAM, DINO, CLIP; built LLM pipelines from scratch and fine-tuned LLaMA; worked with VLMs (LLaVA), reasoning models (LISA).

🎓 EDUCATION

Hong Kong University of Science and Technology (HKUST)

Ph.D., Research focus: Point Cloud Processing, Deep Learning, and BIM Modeling

Hong Kong

Jul 2018 – Aug 2022

Supervisor: Prof. Jack CHENG

Wuhan University

M.Phil., Major: Cartography and GIS

Hubei, China

Sep 2011 – Jun 2013

Supervisor: Prof. Zhongliang Fu

Anhui University of Science and Technology

B.S., Major: Geographic Information System (GIS)

Anhui, China

Sep 2007 – Jun 2011

🏭 RESEARCH EXPERIENCE

Geography Institute of Guangzhou, Guangdong Academy of Sciences

Associate Researcher

Guangzhou, China

Sep 2022 – Present

- Lead research on weakly-supervised 3D point cloud segmentation, long-tailed learning, multi-modal perception, and remote sensing applications.
- Developing large-scale industrial point cloud datasets and benchmarks for MEP scene understanding and safety monitoring.
- Collaborating with **Academician Chenghu Zhou** on low-altitude economy, UAV-based remote sensing, and intelligent urban scene monitoring projects.

Hong Kong University of Science and Technology

Research Assistant at BIM Lab

Hong Kong

Aug 2018 – Aug 2022

- Conducted research on 3D point cloud processing and BIM reconstruction from LiDAR data.
- Developed deep learning methods for automated semantic segmentation of industrial scenes.
- Contributed to Digital Twins for HKUST Campus and scan-to-BIM workflows leading to the ICCBEI 2019 **Best Paper Award**.

- Collaborated with [Prof. Weizhong SHI](#) on smart city and geospatial data analysis.

PUBLICATIONS

Journal Papers (* denote corresponding author)

Under Review / In Progress (5 first-author papers, 2 corresponding-author papers)

- **[First Author]** C. Yin, et al., “Conquering the Dual Crisis: Geometry-Aware Loss for Long-Tailed 3D Point Cloud Segmentation,” *ISPRS Journal of Photogrammetry and Remote Sensing*, under review.
- **[First Author]** C. Yin, et al., “ALPS++: An Efficient Auto-Labeling and Pre-Training Scheme for Remote Sensing Segmentation with Two-Stage Clustering and FeatUp,” *IEEE Transactions on Geoscience and Remote Sensing*, under review.
- **[First Author]** C. Yin, et al., “Enhancing Architectural Style Classification of Chinese Traditional Settlements through Semi-Supervised Deep Learning,” *npj Heritage Science*, under review.
- **[First Author]** C. Yin, et al., “SAM-Assisted Semantic Segmentation of Remote Sensing Images for River Pollution Recognition,” under review.
- **[Corresponding Author]** D. Hu, C. Yin*, “Parameter Analysis and Layout Optimization of a Rooftop Photovoltaic Power System Using a Data-driven Approach,” *Energy*, under review.
- H. Yue, C. Yin, D. Hu, and Z. Guo, “A Deep Learning Framework for Point Cloud Instance Segmentation in Construction Sites,” *Automation in Construction*, under review.
- **[First Author]** C. Yin, et al., “Industrial3D: A Large-Scale, High-Fidelity Dataset and Comprehensive Benchmark for Industrial MEP Scene Understanding,” *Automation in Construction*, *manuscript in preparation*.
- **[Corresponding Author]** Shuju Jing, C. Yin*, “PointBuild-LLM: Multimodal Large Language Model for Indoor Building Component Understanding and Interaction,” *Automation in Construction*, *manuscript in preparation*.

Published Papers (3 first-author Q1 papers, 1 corresponding-author Q2 paper)

- **[First Author]** C. Yin, B. Wang, V. J. L. Gan, M. Wang, and J. C. P. Cheng*, “Automated semantic segmentation of industrial point clouds using ResPointNet++,” *Automation in Construction*, vol. 130, p. 103874, Oct. 2021. (JCR Q1, IF: 10.5)
- **[First Author]** C. Yin, J. C. P. Cheng*, B. Wang, and V. J. L. Gan*, “Automated Classification of Piping Components from 3D LiDAR Point Clouds using SE-PseudoGrid,” *Automation in Construction*, vol. 139, p. 104300, Jul. 2022. (JCR Q1, IF: 10.5)
- **[First Author]** C. Yin, B. Yang, J. C. P. Cheng*, V. J. L. Gan*, B. Wang, and J. Yang, “Label-efficient semantic segmentation of large-scale industrial point clouds using weakly supervised learning,” *Automation in Construction*, vol. 148, p. 104757, Apr. 2023. (JCR Q1, IF: 10.5)
- **[Corresponding Author]** Q. Han, C. Yin*, Y. Deng, and P. Liu, “Towards Classification of Architectural Styles of Chinese Traditional Settlements Using Deep Learning: A Dataset, a New Framework, and Its Interpretability,” *Remote Sensing*, vol. 14, no. 20, Oct. 2022. (JCR Q2, IF: 5.3)
- B. Wang, C. Yin, H. Luo, J. C. P. Cheng, and Q. Wang, “Fully automated generation of parametric BIM for MEP scenes based on terrestrial laser scanning data,” *Automation in Construction*, vol. 125, p. 103615, May 2021. (JCR Q1, IF: 10.5)
- B. Wang, Q. Wang, J. C. P. Cheng, C. Song, and C. Yin, “Vision-assisted BIM reconstruction from 3D LiDAR point clouds for MEP scenes,” *Automation in Construction*, vol. 133, p. 103997, Jan. 2022. (JCR Q1, IF: 10.5)
- B. Wang, Q. Wang*, J. C. P. Cheng*, and C. Yin, “Object verification based on deep learning point feature comparison for scan-to-BIM,” *Automation in Construction*, vol. 142, p. 104515, Oct. 2022. (JCR Q1, IF: 10.5)
- D. Hu, V. J. L. Gan, and C. Yin, “Robot-assisted mobile scanning for automated 3D reconstruction and point cloud semantic segmentation of building interiors,” *Automation in Construction*, vol. 152, p. 104949, Aug. 2023. (JCR Q1, IF: 10.5)
- B. Wang, Z. Chen, M. Li, Q. Wang, C. Yin, and J. C. P. Cheng, “Omni-Scan2BIM: A ready-to-use Scan2BIM approach based on vision foundation models for MEP scenes,” *Automation in Construction*, vol. 162, p. 105384, Jun. 2024. (JCR Q1, IF: 10.5)

Conference Papers

- **[First Author] C. Yin**, W. Boyu, and C. Jack Chin Pang, “Deep learning-based scan-to-BIM framework for complex MEP scene using laser scanning data,” ICCBEI 2019. **Best Paper Award.**
- **[First Author] C. Yin**, J. C. P. Cheng*, B. Wang, and V. J. L. Gan, “Automated Classification of Piping Components from 3D LiDAR Point Clouds using Deep Learning and Squeeze-Excite Mechanism,” IPC 2022.

Book Chapters

- **C. Yin**, B. Wang, K. Chen, H. Liu, C. Huang, J. C. P. Cheng*, and Long Chen, “Digital modelling layer: two digital modelling methods,” In Q. Lu, X. Xie, A.K. Parlikad, J. Schooling and M. Pitt (editors), *Digital Twins in the Built Environment: Fundamentals, principles and applications*, ICE Publishing, 2022, pp. 101–159.

§ RESEARCH FUNDING & PROJECTS

Guangdong Province Overseas Postdoctoral Talent Support Program (PI) Jun 2023 – 2025

- **600K RMB.** Weakly-supervised semantic segmentation for complex indoor 3D perception using point clouds.
- Developing novel deep learning architectures for label-efficient point cloud understanding in complex indoor scenes.

China Postdoctoral Science Foundation (PI) Sep 2023 – Dec 2024

- **80K RMB.** Weakly-supervised semantic segmentation methods for 3D point clouds based on deep long-tail learning.

National Key R&D Program - UAV Cluster Networking for Emergency Rescue 2025

- Proposal writing for key technologies and instant remote sensing applications in urban canyon environments.
- Collaboration with leading institutions on large-scale UAV-based sensing systems.

Xinjiang Talent Development Fund - Low-Altitude Economy Application 2025

- Proposal writing for low-altitude economy project in collaboration with **Academician Chenghu Zhou**.

Space-Air-Ground Integrated Investigation for Land-Based Pollution 2024

- Project implementation and technical reporting for estuarine coastal pollution investigation (Marine Economy Special Project).

National Natural Science Foundation of China (Co-PI) Jan 2021 – Dec 2023

- 300K RMB. Automated identification and extraction of landscape genes for Chinese traditional settlements based on 3D semantic models.

Hong Kong ITF - Automated BIM Generation Oct 2018 – Dec 2021

- 3.4 million HKD. Key participant in automated BIM generation using UAV and indoor 3D laser scanning technologies.

Hong Kong ITF - 3D Geodatabase Framework for Hong Kong Aug 2017 – Jun 2018

- 6.06 million HKD. Key participant in developing 3D Geodatabase Framework for Hong Kong – A Lightweight 3D Seamless Spatial Data Acquisition System (SSDAS).

⚙️ TECHNICAL SKILLS & RESEARCH CAPABILITIES

3D Vision & Point Clouds: ResPointNet++, PointNet++, KPConv, PTV3; semantic/instance segmentation, detection; weakly-supervised, unsupervised and long-tailed learning

Foundation Models & VLMs: SAM series, DINO series, CLIP, LLaVA; adaptation for segmentation and grounding tasks

LLMs & 3D Multi-modal LLMs: LLaMA with **LoRA/QLoRA**; prompt engineering, RAG; VLMs (LLaVA, LISA); 3D M-LLMs (Reason3D, Point-LLM, SegPoint, CityAnchor)

Programming & MLOps: Python (expert, **6+ open-source projects**), PyTorch, TensorFlow, Hugging Face, OpenMMLab, PointCept, Docker, Vim, Git

Geospatial & RS: ArcGIS, QGIS, GDAL, CloudCompare, PCL, FME, Revit; remote sensing workflows

Systems: Linux (Ubuntu/CentOS), CUDA, HPC clusters, Tmux/Screen, Weights & Biases

Academic: LaTeX (expert), paper writing, grant proposals, scientific visualization

PATENTS

An Automatic Annotation and Pre-training Method and System for Remote Sensing Images, *Invention Patent (China)*, Completed. **Chao YIN**, et al. (First Author)

A Semantic Segmentation Method and System for Remote Sensing Images Based on DINO and Parallel Spatial Feature Fusion, *Invention Patent (China)*, Completed. **Chao YIN**, et al. (First Author)

UAV Base Station Deployment Method, Device, Computer Equipment and Storage Medium, *Invention Patent (China)*, Filed (Mar 2025). Jia Sun, Hongda Hu, Shengnan Zhou, **Chao YIN**, etc. (4th Author)

Multimodal Plant Knowledge Query Methods, Systems, and Computer Equipment, *Invention Patent (China)*, Filed (2023). Zhiwei Hou, **Chao YIN**, et al. (2nd Author)

A Parametric 3D Modeling Method for Buildings, *Invention Patent (China)*, Filed (2019). Yuan Liu, Yunyuan Deng, Wenwu Zheng, **Chao Yin**, etc. (4th Author)

TEACHING EXPERIENCE

Teaching Assistant, HKUST (2020-2021): Building Information Modeling and Smart Construction, Infrastructure Systems Engineering

Lecturer, Hengyang Normal University (2013-2017): OOP Programming, GIS Development, Web GIS, GIS System Design

ACADEMIC SERVICE & PROFESSIONAL ACTIVITIES

Peer Reviewer: Automation in Construction, ISPRS Journal of Photogrammetry and Remote Sensing, Geo-spatial Information Science, Results in Engineering, Remote Sensing

Open-Source Contributions: Active maintainer of research code repositories with **140+ GitHub stars**; all published papers have publicly available code for reproducibility

AWARDS & HONORS

Best Paper Award, International Conference on Construction, Building Engineering and Innovation (ICCBEI), 2019

HKUST Postgraduate Scholarship, Full doctoral funding, Hong Kong University of Science and Technology (2018-2022)

First Class Prize in Excellent Graduate Scholarship, Wuhan University (2012)

Last updated: Nov 10, 2025